

FITBUDDY

Project Design Phase

Project: FitBuddy – A Personal Fitness and Wellness Management System

1. Introduction

The design phase converts the requirements of FitBuddy into a clear system structure. It defines the system architecture, modules, database design, user interface, and flow of information. The main goal is to create a simple, secure, and user-friendly fitness management system.

2. System Architecture

- **Presentation Layer:** Mobile or web interface used by users, trainers, and administrators.
- **Application Layer:** Handles login, workout plans, diet tracking, progress tracking, reminders, and business logic.
- **Database Layer:** Stores user profiles, workouts, diet records, progress, and other system data.

3. Main Modules

Module	Main Functions
User Module	Registration, login, profile management
Workout Module	Workout plans, exercise details, daily activity
Diet Module	Meal records, calorie tracking, water intake
Progress Module	Weight, calories, workout duration, progress reports
Notification Module	Workout and healthy-habit reminders
Trainer Module	Workout guidance and user progress monitoring
Admin Module	User, content, and system management

4. Database Design

The database stores and manages the information required by FitBuddy. Important entities include User, Workout, Exercise, Diet, Progress, Trainer, and Notification.

Entity	Important Data
User	User ID, name, email, age, height, weight, goal
Workout	Workout ID, name, type, duration, difficulty
Exercise	Exercise ID, name, instructions, repetitions
Diet	Diet ID, meal type, food, calories, date
Progress	Progress ID, user ID, weight, calories, date
Trainer	Trainer ID, name, contact, specialization

Notification	Notification ID, user ID, message, date, status
--------------	---

5. User Interface Design

- **Login/Register Screen:** Allows users to create an account and sign in.
- **Dashboard:** Displays daily workout, diet, water, and progress information.
- **Workout Screen:** Shows exercise names, instructions, duration, and repetitions.
- **Diet Screen:** Allows users to add and view meals and calorie information.
- **Progress Screen:** Displays fitness progress in a simple format.
- **Profile Screen:** Allows users to view and update personal information.

6. Data Flow Design

User → Login → Dashboard → Select Module → Enter/View Data → Application Processing → Database → Updated Information → User

7. Security Design

- Secure user login and authentication.
- Protect personal and fitness information.
- Use proper access control for users, trainers, and administrators.
- Validate user input before storing data.
- Provide secure database access.

8. Design Principles

- Simple and user-friendly interface.
- Responsive design for different devices.
- Modular system structure for easy maintenance.
- Reliable and accurate data storage.
- Scalable design for future features.

9. Expected Outcome

The design phase provides a complete blueprint for developing FitBuddy. It defines how the modules, database, user interface, and system components work together to provide an effective fitness and wellness management system.